

Project Overview:

This lab focused on identifying active services within a network and implementing defensive controls against common denial-of-service attacks. The assessment provided hands-on experience with reconnaissance, packet analysis, and firewall-based mitigation techniques. The objective was to detect exposed web services, identify malicious ICMP traffic associated with a simulated DDoS event, and evaluate SYN flood mitigation controls within a pfSense firewall environment.

Tools Used:

- Nmap
- Wireshark
- pfSense Firewall
- Kali Linux

Skills Demonstrated:

- Network Reconnaissance
- Service Enumeration
- Packet Analysis
- Firewall Administration
- DDoS Detection

Executive Summary:

The assessment demonstrated how attackers can discover network services and how defenders can detect and mitigate ICMP flood and SYN flood attacks. Nmap was used for reconnaissance, Wireshark for packet analysis, and pfSense for traffic filtering and SYN proxy protection. The project reinforced the importance of continuous monitoring, rate limiting, and firewall tuning.

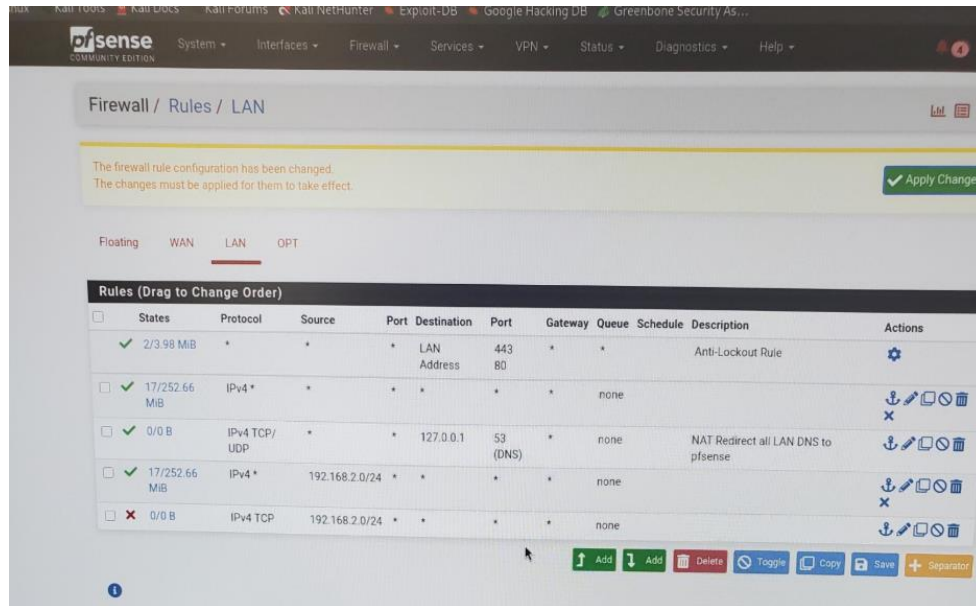
Phase 1: Web Server Discovery:

Used Nmap to identify systems exposing HTTP and HTTPS services.

I used Nmap to scan a specific LAN for open ports 80, 443. The command I used is `Nmap -p 80, 443 192.168.1.0/24`. Nmap was used to scan the local network for TCP ports 80 (HTTP) and 443 (HTTPS). The goal was to identify systems offering web services that could expand the attack surface.

Phase 2: ICMP Flood Detection:

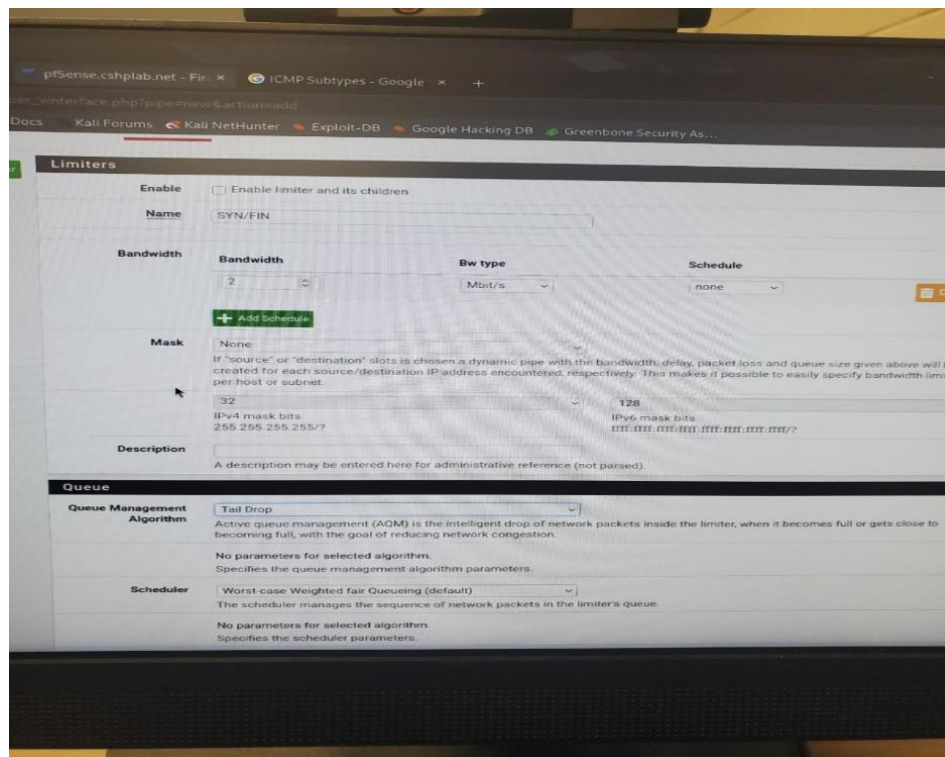
Wireshark was used to analyze packet captures for ICMP Echo Requests, packet volume anomalies, and repeated traffic patterns indicative of flooding behavior. Analyzed ICMP traffic using Wireshark and implemented pfSense controls to mitigate malicious traffic.



Phase 3: SYN Flood Defense:

Evaluated SYN flood behavior and enabled SYN Proxy protections.

A simulated SYN flood generated large numbers of TCP connection requests. The attack was observed to create excessive half-open connections and increase server connection load. I performed a synflood attack by using a randomized Ip. Enabling SYN Proxy protection reduced the impact of incomplete TCP handshakes and improved resistance to spoofed SYN traffic.



Lessons Learned:

The project reinforced the importance of network monitoring, packet analysis, and proactive firewall management.

Conclusion:

This project provided practical experience with network scanning, traffic analysis, and DDoS mitigation.